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BRIEF REPORT

Eye movements are guided by morphological complexity in traditional Mongolian reading

Ming Yan[1,2] · **Ao Min**[3] · **Yaqian Borogjoon Bao**[4] · **Victor Kuperman**[4]

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Abstract

Current research on eye movements in reading has reached a commonly accepted consensus that eye guidance—specifically, the locations of fixations within words—is determined exclusively by low-level visual features. However, this view has been challenged recently by studies in some agglutinative languages, Uighur and Finnish, where saccades have been shown to be influenced also by high-level linguistic features such as morphological complexity. The present study aimed at establishing the generalizability of the effect by extending it to an understudied written language, traditional Mongolian, with a vertical direction of text. Moreover, the current study adopted a corpus-analytic approach, which offers better ecological validity and captures wider ranges of independent variables using much larger datasets than controlled experiments. Consistent with earlier reports, our results demonstrated an influence of morphological complexity on saccades, with first fixations landing closer to the word beginning for morphologically more complex words. The morphological effect was more robust for shorter words and for less frequent words. The results suggest that Mongolian readers can decompose a saccade-target word parafoveally and modulate their saccade execution accordingly.

Keywords Eye movements · Saccade generation · Morphological complexity · Traditional Mongolian

Introduction

During the reading of continuously written texts of sentences and passages, readers send their eye gazes to different positions in the text for lexical processing. Research on eye movements has concluded generally that the decision concerning where to send the gaze for the next fixation is driven by low-level visual features like word length. The behavioral expression of this decision is the first-fixation location (FL), that is, the position where eye gaze initially lands within a word. Typically, the FL distribution peaks near the center of the word (e.g., McConkie et al., 1988; Rayner, 1979). Arguably, such preferred viewing location allows most of the word to be projected onto the foveal retina, taking advantage of the high visual acuity for optimal processing (McConkie et al., 1989; O'Regan & Lévy-Schoen, 1987). One of the ongoing theoretical debates concerns the role of high-level features—that engage deeper cognitive and linguistic processes beyond low-level visual properties such as word shape and length—in parafoveal processing, and how they influence within-word FL. Of special relevance to theories of eye movements in reading is the question whether morphological information, as a form of high-level feature, can be integrated parafoveally. According to serial-attention models of reading, readers encode only low-level visual features in the parafovea. In contrast, parallel lexical processing models assume that various types of information within the perceptual span can be processed simultaneously. Both models can account for many phenomena such as word skipping. They differ, however, with respect to whether parafoveal high-level linguistic features can influence within-word FL. One such feature is morphological complexity of words in the read texts, which can range from simple words (e.g., *bank*) and several types of complex words, including inflected words (*banks* or *banking*), derived words (*banker*)

✉ Ming Yan
mingyan@um.edu.mo

1 Department of Psychology, University of Macau, Macau, China

2 Centre for Cognitive and Brain Sciences, University of Macau, Macau, China

3 School of Mongolian Studies, Inner Mongolia University, Hohhot, China

4 Department of Linguistics and Languages, McMaster University, Hamilton, Canada

or compounds (*banknote*). Whether morphological processing can take place parafoveally has therefore been considered a critical test in the context of the theoretical debate. In the present study, we provide evidence that saccade generation during reading is influenced not only by word length, but also by morphological complexity and word frequency, particularly in the context of an understudied script—vertically written traditional Mongolian.

Empirical evidence has shown clear effects of word frequency and predictability on skipping. Words that are more frequent and more constrained by the preceding context are skipped more often (see Rayner, 2009, for a review). Concerning within-word FL, up until recently, studies on eye movements in reading have suggested a purely low-level eye guidance view. Rayner's (1979) observation of the preferred viewing location effect—the Gaussian distribution of FLs with a peak slightly to the left of the word center—suggests that word length is a primary determinant of saccadic eye movements (see also Nuthmann et al., 2005; Vitu et al., 1990, 2001). The word-based saccade depends critically on accessible information of where each word locates. In most writing systems, low spatial-frequency between words, inter-word spaces, provides exactly such information for word boundary. If inter-word spaces are made unavailable in English texts, saccade strategy changes dramatically. In this case, the beginning of the word takes over as the intended saccade target (Rayner et al., 1998; Rayner & Pollatsek, 1996). Similar patterns have been observed in naturally unspaced scripts such as Japanese (Kajii et al., 2001; Sainio et al., 2007) and Chinese (e.g., Yan et al., 2010), further highlighting the importance of inter-word spaces on eye guidance. Relatedly, little difference in FL has been found between normal reading comprehension and mindless reading, in which readers are required to read sentences with all letters replaced by the letter *z* while preserving inter-word spaces and punctuation (e.g., Nuthmann et al., 2007; Vitu et al., 1995). As such, the saccade planning mechanism can be well preserved, despite the absence of high-level semantic and syntactic information.

Influence of low-level information, other than word length, on eye guidance has also been reported. For instance, Hyönä (1993) hypothesized that infrequent clusters of letters in words could attract readers' eye gaze, because irregular letter strings are visually more salient than regular ones. Supporting the hypothesis, Hyönä (1995) reported that Finnish words with uncommon letter combinations in the first half of the words led to FLs shifting closer to the word beginnings. The influence of orthographic regularity on FL has been demonstrated in a few additional studies (Beauvillain, 1996; Kennedy et al., 2002; Radach et al., 2004). Meanwhile, Hyönä (1995) provided no evidence that the relative informativeness within a word affects FL (Hyönä et al., 1989; Underwood et al., 1990). Relative informativeness refers to the degree to which a part of a word contributes to the identification of the word. For instance, the second half of the word *superstore* (i.e., *store*) is more informative than its first half (i.e., *super*). Overall, studies reviewed above jointly suggest a strong case of the low-level eye guidance view. As a consequence, the low-level guidance of eye movements has been commonly accepted and implemented as one of the core principles of current computational models of eye movements in reading (e.g., Engbert et al., 2005; Reichle et al., 1998).

Importantly in the context of the present study, morphological effects on eye guidance have been previously explored in a number of studies, which, however, have not led to a conclusive outcome. Hyönä and Pollatsek (1998) compared two types of Finnish compound words with the same length and found that the location of the morphological boundary did not affect FL. Deutsch and Rayner (1999) compared morphologically simple (i.e., singular) and complex (i.e., plural) Hebrew words matched on word length, and reported no influence of inflectional morphological constraints on FL. Inhoff et al. (1996) compared English bimorphemic suffixed words and monomorphemic words and also reported an absence of morphological complexity effect on FL, although FL on compound words slightly shifted towards to the word center.

Recently, the notion of low-level eye guidance has been challenged. In particular, some studies have focused on agglutinative languages, Uighur and Finnish, to maximize the chance of observing morphological effects. Arguably, words in these languages are formed by stringing together clearly segmented morphemic units (roots and affixes), each carrying a distinct grammatical or semantic function. Moreover, words in agglutinative languages can be very long and contain multiple morphemes, increasing the likelihood of observing a difference in FL. Yan et al. (2014) reported two studies in Uighur, respectively employing a corpus-analytic approach and an experimental control approach. Their Exp.1 demonstrated that FL shifted closer to the word beginnings for morphologically more complex words, after statistical control for other variables. Experiment 2 manipulated morphological complexity using monomorphemic and double-suffixed target words that were matched closely on word length, part of speech, word frequency, initial bigram frequency, and predictability. Again, FL was closer to the word beginnings for the suffixed words than for the monomorphemic words. Yan et al. provided the first compelling evidence of a morphological complexity effect on within-word FL. Meanwhile, the results imply that the traditional optimal viewing location effect (O'Regan et al., 1984) may be at odd with morphologically complex Uighur words. The deviation from the central fixation pattern suggests that morphological structure plays a significant role in guiding eye movements. Possibly, root morphemes, which may provide information more critical for reading comprehension than suffixes do, attract fixations, echoing the idea that information density within a word can influence saccade generation (Hyönä et al., 1989; Underwood et al., 1990). In a further investigation, Hyönä et al. (2018) adapted the experimental design of Yan et al.'s (2014) Exp. 2 to another agglutinative language, Finnish. Hyönä et al. (2018) also reported FLs closer to the word beginning for the suffixed words than for the monomorphemic words (see also Hyönä et al., 2021). Luo et al. (2023) compared two types of three-character Chinese words that differed in the location of the morphological boundary, which had either a [2 + 1] structure, where the first and the second characters/morphemes formed a more cohesive unit than the second and the third ones, otherwise a [1 + 2] structure. The results showed that FL shifted further away from the word beginnings for the [1 + 2] words than the [2 + 1] words, indicating that saccade may target the longer stems (see also Yan et al., 2025). Together, parafoveal morphological effects on FL have been observed in several orthographies, indicating that high-level information can jointly influence eye guidance.

Does the morphological effect on saccadic generation hold in other languages? Up to now, eye-tracking studies have focused almost exclusively on orthographies with a horizontally rightward direction of text. Several writing systems, such as traditional Chinese, Japanese, and traditional Mongolian, adopt vertical reading and writing. Eye movements during vertical reading also involve two dynamic decisions about where to fixate next and when to move the eyes (e.g., Brysbaert et al., 2005; Vainio et al., 2009), just like horizontal reading. For instance, Yan et al. (2019) reported highly comparable eye movements during horizontal and vertical reading of sentences in traditional Chinese using a within-subject and within-item comparison. In a recent study, Hao and Yan (2025) demonstrated that native Chinese readers, accustomed solely to horizontal reading, exhibited longer fixations and shorter saccades when reading second-language Japanese sentences presented vertically compared to those presented horizontally. This suggests that direction-specific perceptual and oculomotor control experiences can transfer across different writing systems.

As one of the most important factors that influences fixation duration in reading, word frequency effects have been well documented in the existing literature. In general, words that are more frequent are more skipped and are fixated on more briefly than infrequent ones (e.g., Rayner, 1977). However, previous research has shown mixed results regarding the influence of word frequency on FL. While some studies found no word frequency effects (Rayner et al., 1996; White & Liversedge, 2004), other studies indicated that readers' eye gaze landed further into high-frequency words compared to low-frequency ones (Rayner et al., 2006). Additionally, when the initial morphemes of compound words were frequent, FL landed further into these words, regardless of the overall frequency of the compound words (Hyönä & Pollatsek, 1998). In the current study, we aimed to shed new light on the effect of word frequency on FL.

We hypothesized that if the effects of morphological complexity and word frequency on saccade generation are universal across languages, they should be detectable also during vertical reading. Traditional Mongolian is the language of choice in the present study because its distinct linguistic features make it well suited for our research purpose. Mongolian is the official language in Mongolia and the Inner Mongolian Autonomous Region of China. Mongolian utilizes two writing systems: the Cyrillic alphabet and the traditional Mongolian script. While Cyrillic is predominant in Mongolia, the traditional Mongolian script, continuously used since the thirteenth century, remains the predominant writing system in Inner Mongolia. Notably, this script is vertically oriented, written from top to bottom in columns that progress horizontally from left to right across the page. Morphologically, traditional Mongolian exhibits an agglutinative structure characterized primarily by suffixation. In Mongolian, there are only suffixes and no prefixes. Words are systematically formed by attaching suffixes sequentially to lexical roots or stems, encoding grammatical inflection, lexical derivation, and combined grammatical-lexical changes. At its core, traditional Mongolian follows a strict root-plus-suffix hierarchical organization, guided by vowel harmony rules that ensure phonological coherence by mandating suffix forms to align with the root's front/back vowel class. Additionally, the traditional Mongolian script employs positional allography, where each syllable adapts different glyph forms depending on its location—initial, medial, or final—within a word. Overall, these morphological characteristics make traditional Mongolian an ideal candidate for investigating the interplay between morphological complexity and saccadic target selection.

The present study incorporated research ideas reviewed above and aimed at establishing the influence of morphological complexity on FL during the vertical reading of traditional Mongolian texts. The existing evidence is specific to only a few languages and is limited to horizontal reading. Therefore, it is necessary to extend the effect to a different language with a different text orientation. Our prediction is simple: An increase in morphological complexity of the fixated word will come with a shift of the FL towards the beginning of the word.

Method

In this study, we employed the database described by Bao et al. (2025), which extends the Multilingual Eye Movement Corpus (MECO; Siegelman et al., 2022) to include vertically written traditional Mongolian. For ease of reference, we repeat the key points below.

Participants

The database includes eye-movement data from 68 participants (53 women; age = 20.82 ± 2.15 years, range: 18–27 years) from Inner Mongolia Normal University in Hohhot, China. All participants were native speakers of Mongolian and attended Mongolian schools from kindergarten through university, where they studied all subjects in Mongolian. This extensive education made them proficient in reading vertically written traditional Mongolian. Their self-ratings of Mongolian proficiency (on a scale of 0–10) were as follows: speaking (mean = 9.03 ± 0.69), oral comprehension (mean = 9.10 ± 0.76), and reading (mean = 9.16 ± 0.64). In addition to their native language, all participants were proficient in Chinese, which they began learning in the third year of elementary school. Each participant had normal or corrected-to-normal eyesight. The participants also completed a nonverbal IQ test using the short version of the Culture Fair Test-3 (CFT20), Subset 3 Matrices, Form A19, to provide a standardized measure of nonverbal intelligence.

Material

Although traditional Mongolian is an alphabetic script, it functions syllabically in practice. Consonants and vowels do not form words letter by letter; instead, they combine into syllabic units. Each unit typically consists of a consonant–vowel combination or a standalone vowel, and these syllabic units are the basic building blocks of written words. Furthermore, the visual form of each syllabic unit varies depending on its position within a word—initial, medial, or final—resulting in syllables of different visual lengths. Consequently, words are segmented and perceived by syllabic units rather than individual letters.

As mentioned earlier, the traditional Mongolian writing system is an ideal example to explore morphological effects due to its rich suffix system. Derivational and inflectional suffixes follow this positional adaptation, integrating seamlessly with root forms into unified orthographic words. For instance, the verb *bichi* (“write”) can become the noun *bichig* (*bichi-g*, “writing, text”) through the nominalizing suffix *-g* and further derive into another noun *bichigten* (*bichi-g-ten*, “writer”) by adding the suffix *-ten*, or become the verb *bichigdeg* (*bichi-g-deg*, “being written”) by adding the suffix *-deg*, or even derive the more complex word *bichigdegsen* (*bichi-g-deg-sen*, “that which has been written”). Verbal inflection occurs through suffixation as well, as illustrated by *bichi* (“write”) becoming *bichibe* (*bichi-be*, “wrote”) with the past-tense suffix *-be*, or *bichine* (*bichi-ne*, “will write”) with the future-tense suffix *-ne*. Nominal inflection is achieved primarily through a rich system of grammatical cases. Unlike verbal suffixes, nominal case markers appear as separate orthographic units rather than joining the root noun. For instance, the genitive case *on*, indicating possession, is written separately as *bichig on* (“of the book”), not combined as *bichigon*; similarly, the ablative case marker *es* is written as *bichig es* (“from the book”), not combined as *bichiges*; pluralization is achieved by adding suffixes such as *ud*, changing *bichig* (“book”) to *bichig ud* (“books”). In Mongolian, suffixes are counted based on their distinct morphological roles and how they are sequentially attached to a root or stem. Each suffix that adds a unique grammatical or semantic function is considered a separate affix. These suffixes are identified by analyzing the word’s structure and recognizing each morpheme that modifies the base form. In the current study, we did not distinguish between derivational and inflectional suffixes, because previous studies have shown that they have similar influences on FL (e.g., Yan et al., 2014).

The database contains 12 traditional Mongolian texts comprising 99 sentences and 2,592 words. Among them, 1,669 (64.4%) contain no suffixes, 558 (21.5%) contain one suffix, 282 (10.9%) contain two suffixes, 74 (2.9%) contain three suffixes, eight (0.3%) contain four suffixes, and one word (0.04%) contains five suffixes. All passages were converted to images and were presented using Experiment Builder (Version 2.4.1, SR Research, Kanata, Ontario, Canada). Figure 1 illustrates an example of the experimental materials.

Apparatus

Eye movements were recorded using the Tower Mount EyeLink 1000 eye tracker (SR Research, Kanata, Ontario, Canada) at a sampling rate of 1000 Hz. The stimuli were displayed on a 20-inch Lenovo L2021 monitor with a resolution of 1,024 × 768 pixels. The texts were printed in Menk Qagan Tig font (font size: 19 points) with 1.5 spacing. Participants were seated 54 cm away from the monitor and used a chin rest and head restraint to minimize head movements. As with other languages that cannot be typed with monospaced fonts due to the inherently different length of letters (e.g., those using the Arabic script), measurement of the visual angle in orthographic units is complicated. In our data, an average syllable subtended approximately 1° of the visual angle (Borjigin et al., 2024). They read sentences binocularly, but only their right eye was monitored.

Procedure

Before starting the eye-tracking experiment session, each participant completed a basic demographic and language proficiency questionnaire, as well as the nonverbal CFT20

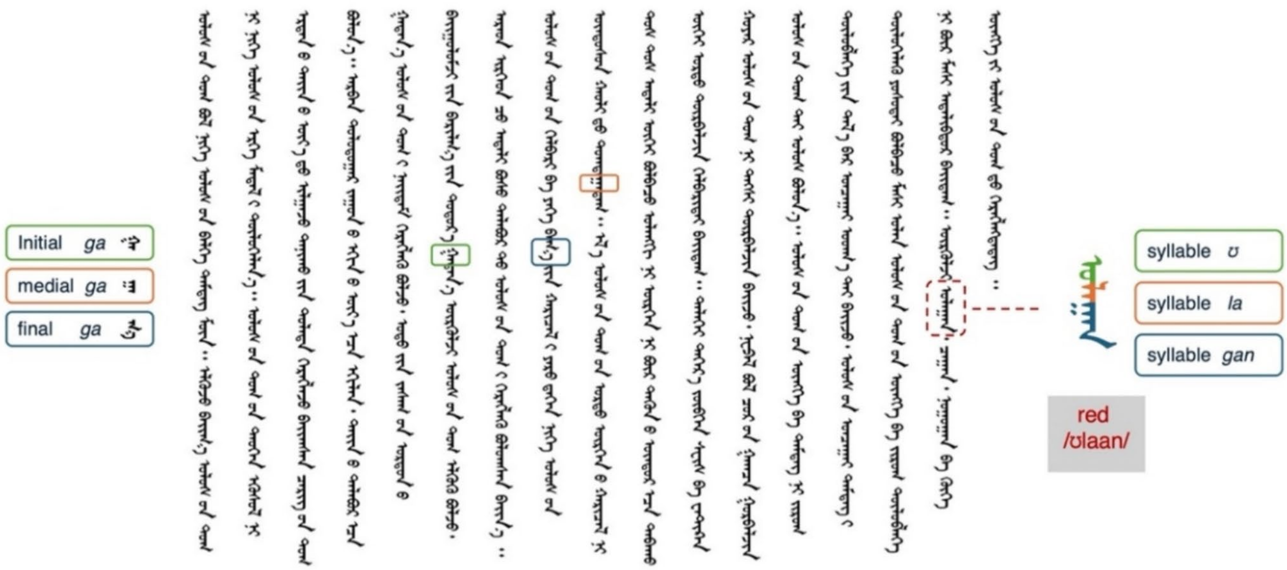

Fig. 1 Example material. *Note.* The syllable/ga/is shown in its initial, medial, and final forms to demonstrate positional allography in traditional Mongolian script. An excerpt from the experimental sentence highlights one occurrence of the/ga/syllable. On the right, the word for "red" (/ʊlaan/) is decomposed into three syllabic units:/ʊ/,/la/, and/gan/, each marked by its respective visual form. This exemplifies how words in traditional Mongolian are composed of syllabic units that vary in shape depending on their position within the word. (Color figure online)

test. Afterward, the experimenter guided them through setting up the eye-tracking session and conducted a nine-point calibration and validation. Participants were then introduced to the experiment with one practice text, followed by four yes/no questions displayed one at a time. Participants pressed "1" on the keyboard for "yes" and "0" for "no" to answer the questions. Once it was confirmed that the participants understood the task, the formal experiment began in following the same procedure. Before each trial, a fixation dot appeared on the monitor slightly above the first word of the passage. The trial began once the participant fixated on the dot. This drift check and correction procedure occurred at the beginning of each trial, with calibration monitored by the experimenter and redone as needed. Each of the 12 texts was presented on a separate screen. Participants were instructed to read the passages silently for comprehension and press the space bar once they finished reading each passage.

Data analysis

Trials containing tracker errors or participants' blinks or coughs during reading were excluded from the analyses ($n = 11$). Additionally, the first fixated words in each line of text were discarded, because FLs on these words do not reflect typical saccade generation and often are outcomes of the return-sweeps from previous lines. Words with first-fixation durations (duration of the first fixation on a word, irrespective of total number of fixations) shorter than 60 ms or longer than 800 ms and gaze durations (the sum of all fixation durations during the first-pass reading of a word) longer than 1,200 ms were removed ($n = 5{,}208$, 2.8%). Together we kept 67,718 first-pass observations of FL for data analysis. Given the nature of the writing system which makes it difficult to determine the boundaries of a single character, FL was calculated by taking the distance between the vertical position of the initial fixation on a word and the word's starting point on the screen (y-coordinate), then dividing this by the word's length. Thus, FL reflected the landing position of the first fixation relative to the word length.

Estimates are based on linear mixed models (LMMs) using the *lme4* package (Version 1.1–23; Bates et al., 2015) and the *lmerTest* package (Version 3.1–2; Kuznetsova et al., 2017) in the R environment (Version 4.4.1; R Development Core Team, 2024). The partial effect reported below reflects the unique contribution of a given predictor, isolated from potential confounding influences. Statistically speaking, the partial effect of a predictor is represented by its coefficient estimate, which quantifies the expected change in the dependent variable per unit change in the predictor, while holding all other predictors constant. The fixed effects included word length, log-transformed word frequency, morphological complexity, and all their two-way interactions. Generally, launch site—the distance between the location of the last fixation and the beginning of the currently fixated word—cannot be controlled prior to the experiment and has shown robust influences on eye-movement parameters. The relationship between launch site and FL reflects both linear and quadratic influences (McConkie et al., 1988). As the launch site increases, fixations tend to land closer to the beginning of the word. Additionally, fixations originating from intermediate launch sites are more likely to land near the word center, whereas those from very close or very distant launch sites tend to be less accurate, resulting in a broader distribution of FL. To capture such influences, linear and quadratic trends of launch site were included as model covariates for statistical control. All continuous predictors were centered before entering the model. We included subject- and item-related variance components for the intercepts and random-slopes for the fixed-effects and started with full random-effects. Following a standard parsimonious LMMs selection procedure (Matuschek et al., 2017), we dropped correlation parameters and small variance parameters for successful model convergence.

Results

The FL distribution overall formed a Gaussian distribution with a peak between the word beginning and the center (Fig. 2), which highly resembles those from horizontal reading (e.g., Nuthmann et al., 2005; Rayner, 1979; Vitu et al., 2001; Yan et al., 2014). Therefore, the saccade generation mechanism of vertical reading is largely compatible with that of horizontal reading.

Table 1 presents the model output. The largest effects were launch site and word length. FL decreased with increases of launch site and word length. The launch site

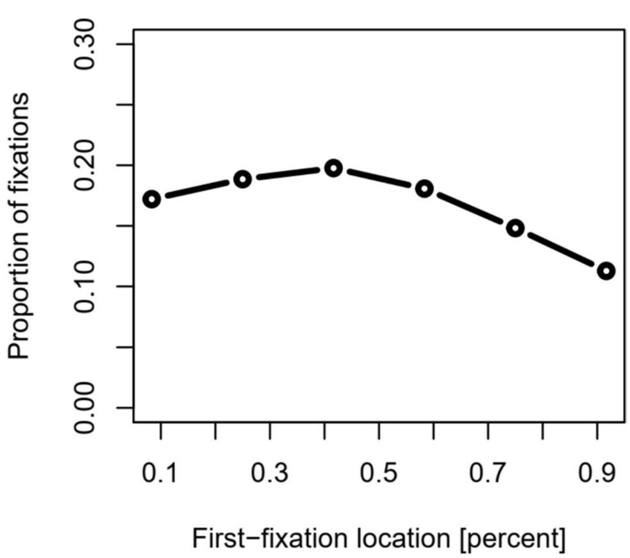

Fig. 2 Fixation location distribution

Table 1 Linear mixed-model estimates of first-fixation location

Fixed effect	Estimate	Std. Error	*t* value	*p* value
(Intercept)	0.475	0.006	78.366	< 0.001
Poly (LS, 2)1	− 21.26	0.260	− 81.653	< 0.001
Poly (LS, 2)2	12.85	0.253	50.729	< 0.001
LEN	− 0.018	0.001	− 18.414	< 0.001
LWF	0.022	0.007	3.025	0.002
AFX	0.006	0.006	1.061	0.289
LEN × LWF	0.009	0.002	4.412	< 0.001
LEN × AFX	0.002	0.001	3.269	0.001
LWF × AFX	0.044	0.017	2.520	0.011

LS = launch site, LEN = word length, LWF = log word frequency, AFX = number of affixes

effect on FL is a canonical oculomotor effect and has been reported consistently across different writing systems. As word length increases, FL is known to shift more to the left of the word center (McConkie et al., 1988; Rayner, 1979). These effects are not of our primary interest; however, the fact that they replicate previous findings suggests that the current data are reliable.

Given the low-to-moderate correlations between predictors, we checked the degree of multicollinearity in our model, using the Variance Inflation Factor (VIF). The rule of thumb is that VIF values above 10 signal unacceptable level of collinearity, requiring removal or transformation of variables. All VIF values in our data were in the 1–5 range, represent moderate collinearity which is not a cause for concern.

The theoretically important effects discovered here are significant interactions between morphological complexity and word length, between morphological complexity and word frequency, between word length and frequency, and the significant main effect of word frequency. Although the main effect of morphological complexity was not significant by itself, there was an interaction with word length, indicating the influence of morphological complexity on FL was more robust for shorter words (Fig. 3, top panel). Presumably, parafoveal morphological complexity information is difficult to obtain when the word is very long. Meanwhile, the interaction with word frequency revealed a reliable morphological effect only for low-frequency words (Fig. 3, bottom panel). High-frequency words tend to be morphologically simpler than low-frequency ones ($r = -0.360$; $p < 0.01$), which may undermine parafoveal morphological decomposition. The interaction between word length and frequency indicated that the influence of word frequency on FL was more pronounced for longer words, possibly because shorter words offer less room for variation. Finally, the main effect of word frequency showed a rightward shift of FL for higher-frequency words after statistically controlling for word length.

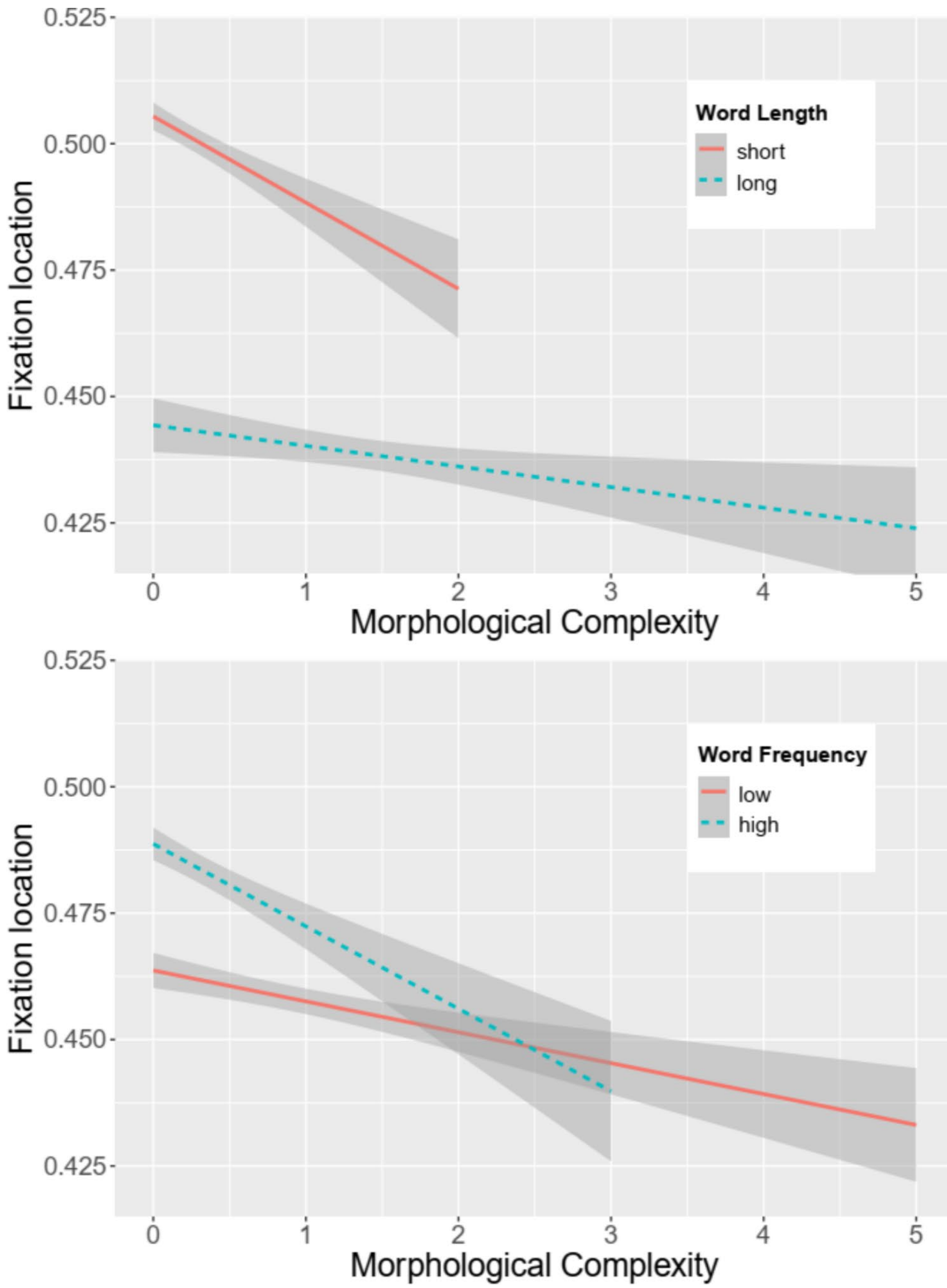

Fig. 3 Morphological complexity effects on first-fixation location. *Note.* Interactions between morphological complexity and word length (top panel), and between morphological complexity and word frequency (bottom panel) on first-fixation location. Short words are defined as those containing six or fewer letters. Using a quantile-based cut, low-frequency words are defined as those with a log-frequency value below − 0.293. This results in 33,967 observations of high-frequency words and 33,761 observations of low-frequency words. Lines and shaded 95% confidence intervals are partial effects based on linear mixed model, generated using the *remef* package (Version 0.6.10; Hohenstein & Kliegl, 2015) and the *ggplot2* package (Version 2.0.0; Wickham, 2009). (Color figure online)

We conducted an additional analysis focusing exclusively on content words based on 59,200 observations, given the very limited number of function words in the language ($n = 14$). As expected, this analysis replicated the key pattern of results. Significant effects were observed for the linear and quadratic trends of launch site ($b = -21.93$, $SE = 0.274$, $t = -80.030$, $p < 0.001$ and $b = 13.08$, $SE = 0.263$, $t = 49.755$, $p < 0.001$), word length ($b = -0.017$, $SE = 0.001$, $t = -12.777$, $p < 0.001$), and word frequency ($b = 0.040$, $SE = 0.011$, $t = 3.696$, $p < 0.001$). More critically, the interactions between word length and frequency ($b = 0.014$, $SE = 0.003$, $t = 4.317$, $p < 0.001$), and between morphological complexity and word length ($b = 0.002$, $SE = 0.001$, $t = 3.061$, $p = 0.002$), remained significant.

Discussion

The present study aimed to understand eye guidance during the vertical reading of traditional Mongolian text and document language universal influences on saccadic eye movements. The results confirmed a number of robust findings that has been observed consistently in previous studies. For instance, we found that the FLs formed a Gaussian distribution with its peak located between the beginning and the center of the word, suggesting the primary role of word length in saccade-target selection (Rayner, 1979). The underlying mechanism of the FL distribution likely reflects that readers allocate more attention to the upcoming text in the direction of reading to acquire upcoming information from unprocessed words (McConkie & Rayner, 1975). Therefore, our results agree with other eye-tracking studies on vertical reading (Borjigin et al., 2024; Su et al., 2020; Yan et al., 2019, 2024). Meanwhile, FL landed further into high-frequency words than low-frequency ones, replicating previous reports (e.g., Rayner et al., 2006). The most critical contribution we made in the present study is providing new evidence on the role of morphological complexity in eye guidance. Below we discuss three inter-related aspects of our findings, including serial and parallel models of eye movements, morphological complexity effects on FL in other languages, and corpus analytic and experimental control approaches of eye-movement research.

Current computational models of eye movements in reading can be divided into two large categories. Sequential attention shift models like E-Z Reader (Reichle et al., 1998) propose that lexical processing happens in a strictly linear fashion, focusing only on the word currently being attended to. Attention can move to the next word only after the current word has been fully processed. In contrast, attentional gradient models, such as SWIFT (Engbert et al., 2005), Glenmore (Reilly & Radach, 2003), and OB1-Reader (Snell et al., 2018), suggest that lexical processing is distributed across multiple words within the perceptual span. The two types of models share many common mechanisms. For instance, word skipping effects are generally compatible with these models and have been incorporated similarly. Considering their basic principles, different predictions can be derived. In general, parafoveal processes of high-level information, such as semantics and morphology, are generally not expected in the serial models, whereas these processes are in line with the parallel models. Although both types of models advocate that low-level information guides within-word FL, the effects of morphological complexity on FL appear to be more compatible with the parallel models, because the basic assumption of distributed attention and more probabilistic saccade targeting allow the models for more flexibility in saccadic targeting. In particular, the Glenmore model (Reilly & Radach, 2006) uses a saliency map that integrates visual features (e.g., word length and contrast) and lexical features (e.g., word frequency and predictability) for saccade targeting. Therefore, it might have the best chance to capture the morphological influences on FL. As such, our results provide critical theoretical insights for future developments of computational models. Although the morphological effects align with the core principle of parallel models, accommodating this property remains a challenge, as it may require implementing dynamic parafoveal morphological decomposition processes.

The morphological effect is theoretically relevant to one fundamental debate on whether high-level linguistic information can influence where the eyes initially fixate on within a word (see Rayner, 2009, for a review). Classic research on eye guidance has reached a widely accepted conclusion that readers select the center of the next word as the intended FL and execute a saccade towards it for optimal lexical processing based on word length (McConkie et al., 1989). Word length can be inferred from spatial frequency information in the parafoveal and peripheral vision. In addition, early studies have failed to find evidence for parafoveal processing of morphological information (Kambe, 2004; Lima, 1987). This low-level eye guidance view has been challenged, for the first time, by corpus analysis of eye movements. Yan et al. (2014) demonstrated that with an increase in morphological complexity, FL shifted further towards the beginning of the word, providing evidence that eye guidance can be influenced by parafoveal morphological decomposition, at least in Uighur. Follow-up experiments comparing monomorphemic and double-suffixed target words confirmed the morphological effect (Hyönä et al., 2018; Yan et al., 2014, Exp. 2). The present study is the first direct replication of the morphological complexity effect on FL from a corpus analytic perspective.

Yan et al. (2014) proposed two mechanisms for the morphological effect on FL. First, it can be more challenging to identify the morphological boundaries of suffixed words compared to monomorphemic words. An increase in processing difficulty results in shorter saccades, causing FLs to shift closer to the beginning of the word. Second, readers might focus on the word stem (see also Grainger & Beyersman, 2017), which is shorter for suffixed words, also leading FLs to fall closer to the beginning of the word. Hyönä et al. (2021) proposed a third possibility that the morphological effect lies in perceiving word-ending suffixes as highly frequent letter clusters (e.g., Inhoff, 1989; Johnson et al., 2007). These suffixes are identifiable even at eccentric locations. In this case, FL shifting away from the suffixes (and closer to the stems) promotes overall word recognition. While the present study has extended the morphological effect to a new orthography, the underlying nature of the morphological effect remains to be established.

Methodologically, most studies on morphological effects on eye guidance adopt an experimental-control approach. Nevertheless, the statistical-control approach has its unique advantages. Corpus analysis, which uses larger datasets and covers much wider ranges of model predictors, offers better ecological validity. It can reveal subtle effects and patterns that controlled experiments might miss due to their limited conditions (Risse et al., 2008). For instance, the Multilingual Eye-movement Corpus (MECO) project (Siegelman et al., 2022), which has adopted such a statistical-control approach and compares multiple languages through eye-tracking data during natural passage reading, may offer insights that are difficult to obtain in a multilingual factorial experiment, due to the wide diversity of the written languages under study.

In conclusion, the processing of high-level information in the parafovea remains a central and actively debated issue in the field of eye movement research. Evidence that morphological structure guides within-word FL has so far been demonstrated only in two horizontal reading scripts. The vertical reading of Mongolian passages in the current study provides a valuable opportunity for both consolidation and extension of this line of research, supporting the notion of a language-universal morphological influence on saccade generation.

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Data availability The data and scripts for analyses are publicly available at https://osf.io/HC5XD/ and the experiment was not preregistered.

Code availability See above.

Declarations

Ethics approval The experimental procedures were approved by the Ethics Committee of Inner Mongolia Normal University.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication Participants provided informed consent for publication of their data.

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

References

Bates, D., Maechler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software, 67*, 1–48. https://doi.org/10.18637/jss.v067.i01

Bao, Y., Li, X., & Kuperman, V. (2025). The eye movement database of passage reading in vertically written traditional Mongolian. *Scientific Data, 12*, 499. https://doi.org/10.1038/s41597-025-04771-w

Beauvillain, C. (1996). The integration of morphological and whole-word form information during eye fixations on prefixed and suffixed words. *Journal of Memory and Language, 35*(6), 801–820. https://doi.org/10.1006/jmla.1996.0041

Borjigin, B., Zhang, G., Hou, Y., & Li, X. (2024). Perceptual span in Mongolian text reading. *Current Psychology, 43*(29), 24287–24294. https://doi.org/10.1007/s12144-024-06074-6

Brysbaert, M., Drieghe, D., & Vitu, F. (2005). Word skipping: Implications for theories of eye movement control in reading. In G. Underwood (Ed.), *Cognitive processes in eye guidance* (pp. 53–78). Oxford University Press. https://doi.org/10.1093/acprof:oso/9780198566816.003.0003

Deutsch, A., & Rayner, K. (1999). Initial fixation location effects in reading Hebrew words. *Language and Cognitive Processes, 14*(4), 393–421. https://doi.org/10.1080/016909699386284

Engbert, R., Nuthmann, A., Richter, E. M., & Kliegl, R. (2005). SWIFT: A dynamical model of saccade generation during reading. *Psychological Review, 112*(4), 777–813. https://doi.org/10.1037/0033-295X.112.4.777

Grainger, J., & Beyersman, E. (2017). Edge-aligned embedded word activation initiates morpho-orthographic segmentation. In B. H. Ross (Ed.), *The psychology of learning and motivation* (pp. 285–317). Elsevier Academic. https://doi.org/10.1016/bs.plm.2017.03.009

Hao, Y., & Yan, M. (2025). Is direction-specific reading experience transferrable across orthographies? Eye tracking evidence from Chinese-Japanese bilinguals. *Scientific Studies of Reading, 29*(3), 290–302. https://doi.org/10.1080/10888438.2025.2472631

Hohenstein, S., & Kliegl, R. (2015). *remef: Remove partial effects* [Computer software]. https://github.com/hohenstein/remef/

Hyönä, J. (1993). Effects of thematic and lexical priming on readers' eye movements. *Scandinavian Journal of Psychology, 34*(4), 293–304. https://doi.org/10.1111/j.1467-9450.1993.tb01126.x

Hyönä, J. (1995). Do irregular letter combinations attract readers' attention? Evidence from fixation locations in words. *Journal of Experimental Psychology: Human Perception and Performance, 21*(1), 68–81. https://doi.org/10.1037/0096-1523.21.1.68

Hyönä, J., Heikilä, T. T., Vainio, S., & Kliegl, R. (2021). Parafoveal access to word stem during reading: An eye movement study. *Cognition, 208*, 104547. https://doi.org/10.1016/j.cognition.2020.104547

Hyönä, J., Niemi, P., & Underwood, G. (1989). Reading long words embedded in sentences: Informativeness of word halves affects eye movements. *Journal of Experimental Psychology: Human Perception and Performance, 15*(1), 142–152. https://doi.org/10.1037/0096-1523.15.1.142

Hyönä, J., & Pollatsek, A. (1998). Reading finnish compound words: Eye fixations are affected by component morphemes. *Journal of*

Experimental Psychology: Human Perception and Performance, 24(6), 1612–1627. https://doi.org/10.1037/0096-1523.24.6.1612
Hyönä, J., Yan, M., & Vainio, S. (2018). Morphological structure influences the initial landing position in words during reading Finnish. *The Quarterly Journal of Experimental Psychology, 71*(1), 122–130. https://doi.org/10.1080/17470218.2016.1267233
Inhoff, A. W. (1989). Parafoveal processing of words and saccade computation during eye fixations in reading. *Journal of Experimental Psychology: Human Perception and Performance, 15*, 544–555. https://doi.org/10.1037/0096-1523.15.3.544
Inhoff, A. W., Briihl, D., & Schwartz, J. (1996). Compound word effects differ in reading, on-line naming, and delayed naming tasks. *Memory & Cognition, 24*(4), 466–476. https://doi.org/10.3758/BF03200935
Johnson, R. L., Perea, M., & Rayner, K. (2007). Transposed-letter effects in reading: Evidence from eye movements and parafoveal preview. *Journal of Experimental Psychology: Human Perception and Performance, 33*, 209–229. https://doi.org/10.1037/0096-1523.33.1.209
Kambe, G. (2004). Parafoveal processing of prefixed words during eye fixations in reading: Evidence against morphological influences on parafoveal preprocessing. *Perception & Psychophysics, 66*(2), 279–292. https://doi.org/10.3758/BF03194879
Kajii, N., Nazir, T. A., & Osaka, N. (2001). Eye movement control in reading unspaced text: The case of the Japanese script. *Vision Research, 41*(19), 2503–2510. https://doi.org/10.1016/S0042-6989(01)00132-8
Kennedy, A., Pynte, J., & Ducrot, S. (2002). Parafoveal-on-foveal interactions in word recognition. *The Quarterly Journal of Experimental Psychology a: Human Experimental Psychology, 55A*(4), 1307–1337. https://doi.org/10.1080/02724980244000071
Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. (2017). LmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software, 82*(13), 1–26. https://doi.org/10.18637/jss.v082.i13
Lima, S. D. (1987). Morphological analysis in sentence reading. *Journal of Memory and Language, 26*(1), 84–99. https://doi.org/10.1016/0749-596X(87)90064-7
Luo, Y., Tan, D., & Yan, M. (2023). Morphological structure influences saccade generation in Chinese reading. *Reading and Writing, 36*(5), 1339–1355. https://doi.org/10.1007/s11145-022-10325-y
Matuschek, H., Kliegl, R., Vasishth, S., Baayen, H., & Bates, D. (2017). Balancing type I error and power in linear mixed models. *Journal of Memory and Language, 94*, 305–315. https://doi.org/10.1016/j.jml.2017.01.001
McConkie, G. W., Kerr, P. W., Reddix, M. D., & Zola, D. (1988). Eye movement control during reading: I. The location of initial eye fixations on words. *Vision Research, 28*(10), 1107–1118. https://doi.org/10.1016/0042-6989(88)90137-X
McConkie, G. W., Kerr, P. W., Reddix, M. D., Zola, D., & Jacobs, A. M. (1989). Eye movement control during reading: II. Frequency of refixating a word. *Perception & Psychophysics, 46*, 245–253. https://doi.org/10.3758/BF03208086
McConkie, G. W., & Rayner, K. (1975). The span of the effective stimulus during a fixation in reading. *Perception & Psychophysics, 17*, 578–586. https://doi.org/10.3758/BF03203972
Nuthmann, A., Engbert, R., & Kliegl, R. (2005). Mislocated fixations during reading and the inverted optimal viewing position effect. *Vision Research, 45*(17), 2201–2217. https://doi.org/10.1016/j.visres.2005.02.014
Nuthmann, A., Engbert, R., & Kliegl, R. (2007). The IOVP effect in mindless reading: Experiment and modeling. *Vision Research, 47*(7), 990–1002. https://doi.org/10.1016/j.visres.2006.11.005
O'Regan, J. K., Lévy-Schoen, A., Pynte, J., & Brugaillère, B. (1984). Convenient fixation location within isolated words of different length and structure. *Journal of Experimental Psychology: Human Perception and Performance, 10*(2), 250–257. https://doi.org/10.1037/0096-1523.10.2.250
O'Regan, J. K., & Lévy-Schoen, A. (1987). Eye-movement strategy and tactics in word recognition and reading. In M. Coltheart (Ed.), *Attention and performance: The psychology of reading* (pp. 363–383). Erlbaum.
R Development Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing.
Radach, R., Inhoff, A., & Heller, D. (2004). Orthographic regularity gradually modulates saccade amplitudes in reading. *European Journal of Cognitive Psychology, 16*, 27–51. https://doi.org/10.1080/09541440340000222
Rayner, K. (1977). Visual attention in reading: Eye movements reflect cognitive processes. *Memory & Cognition, 5*(4), 443–448. https://doi.org/10.3758/BF03197383
Rayner, K. (1979). Eye guidance in reading: Fixation locations within words. *Perception & Psychophysics, 8*, 21–30. https://doi.org/10.1068/p080021
Rayner, K. (2009). Eye movements and attention in reading, scene perception, and visual search. *Quarterly Journal of Experimental Psychology, 62*, 1457–1506. https://doi.org/10.1080/17470210902816461
Rayner, K., Fischer, M. H., & Pollatsek, A. (1998). Unspaced text interferes with both word identification and eye movement control. *VisionResearch, 38*(8), 1129–1144. https://doi.org/10.1016/S0042-6989(97)00274-5
Rayner, K., & Pollatsek, A. (1996). Reading unspaced text is not easy: Comments on the implications of Epelboim et al.'s (1994) study for models of eye movement control in reading. *Vision Research, 36*(3), 461–465. https://doi.org/10.1016/0042-6989(95)00132-8
Rayner, K., Reichle, E. D., Stroud, M. J., Williams, C. C., & Pollatsek, A. (2006). The effect of word frequency, word predictability, and font difficulty on the eye movements of young and older readers. *Psychology and Aging, 21*, 448–465. https://doi.org/10.1037/0882-7974.21.3.448
Rayner, K., Sereno, S. C., & Raney, G. E. (1996). Eye movement control in reading: A comparison of two types of models. *Journal of Experimental Psychology: Human Perception and Performance, 22*(5), 1188–1200. https://doi.org/10.1037/0096-1523.22.5.1188
Reichle, E. D., Pollatsek, A., Fisher, D. L., & Rayner, K. (1998). Toward a model of eye movement control in reading. *Psychological Review, 105*, 125–157. https://doi.org/10.1037/0033-295X.105.1.125
Reilly, R. G., & Radach, R. (2003). Glenmore: An interactive activation model of eye movement control in reading. In J. Hyönä, R. Radach, & H. Deubel (Eds.), *The mind's eye: Cognition and applied aspects of eye movement research* (pp. 429–456). Elsevier.
Reilly, R. G., & Radach, R. (2006). Some empirical tests of an interactive activation model of eye movement control in reading. *Cognitive Systems Research, 7*(1), 34–55. https://doi.org/10.1016/j.cogsys.2005.07.006
Risse, S., Engbert, R., & Kliegl, R. (2008). Eye movement control in reading: Experimental and corpus-analysis challenges for a computational model. In K. Rayner, D. Shen, X. Bai, & G. Yan (Eds.), *Cognitive and cultural influences on eye movements* (pp. 65–91). Tianjin People's Publishing House.
Sainio, M., Hyönä, J., Bingushi, K., & Bertram, R. (2007). The role of interword spacing in reading Japanese: An eye movement study. *Vision Research, 47*(20), 2575–2584. https://doi.org/10.1016/j.visres.2007.05.017
Siegelman, N., Schroeder, S., Acartürk, C., Ahn, H.-D., Alexeeva, S., Amenta, S., Bertram, R., Bonandrini, R., Brysbaert, M., Chernova, D., Da Fonseca, S. M., Dirix, N., Duyck, W., Fella, A., Frost, R., Gattei, C. A., Kalaitzi, A., Kwon, N., Lõo, K., … Kuperman, V. (2022). Expanding horizons of cross-linguistic research on reading: The multilingual eye-movement corpus (MECO).

Behavior Research Methods, 54, 2843–2863. https://doi.org/10.3758/s13428-021-01772-6

Snell, J., van Leipsig, S., Grainger, J., & Meeter, M. (2018). OB1-reader: A model of word recognition and eye movements in text reading. *Psychological Review, 125*(6), 969–984. https://doi.org/10.1037/rev0000119

Su, J., Yin, G., Bai, X., Yan, G., Kurtev, S., Warrington, K. L., & Paterson, K. B. (2020). Flexibility in the perceptual span during reading: Evidence from Mongolian. *Attention, Perception, & Psychophysics, 82*(4), 1566–1572. https://doi.org/10.3758/s13414-019-01960-9

Underwood, G., Clews, S., & Everatt, J. (1990). How do readers know where to look next? Local information distributions influence eye fixations. *Quarterly Journal of Experimental Psychology, 42*(1), 39–65. https://doi.org/10.1080/14640749008401207

Vainio, S., Hyönä, J., & Pajunen, A. (2009). Lexical predictability exerts robust effects on fixation duration, but not on initial landing position during reading. *Experimental Psychology, 56*, 66–74. https://doi.org/10.1027/1618-3169.56.1.66

Vitu, F., McConkie, G. W., Kerr, P., & O'Regan, J. K. (2001). Fixation location effects on fixation durations during reading: An inverted optimal viewing position effect. *Vision Research, 41*(25/26), 3513–3533. https://doi.org/10.1016/S0042-6989(01)00166-3

Vitu, F., O'Regan, J. K., Inhoff, A. W., & Topolski, R. (1995). Mindless reading: Eye-movement characteristics are similar in scanning letter strings and reading texts. *Perception & Psychophysics, 57*(3), 352–364. https://doi.org/10.3758/BF03213060

Vitu, F., O'Regan, J. K., & Mittau, M. (1990). Optimal landing position in reading isolated words and continuous text. *Perception & Psychophysics, 47*(6), 583–600. https://doi.org/10.3758/BF03203111

White, S. J., & Liversedge, S. P. (2004). Orthographic familiarity influences initial eye fixation positions in reading. *European Journal of Cognitive Psychology, 16*(1/2), 52–78. https://doi.org/10.1080/09541440340000204

Wickham, H. (2009). *Ggplot2: Elegant graphics for data analysis*. Springer.

Yan, M., Kliegl, R., & Pan, J. (2024). Direction-specific reading experience shapes perceptual span. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 50*(11), 1740–1748. https://doi.org/10.1037/xlm0001340

Yan, M., Kliegl, R., Richter, E. M., Nuthmann, A., & Shu, H. (2010). Flexible saccade-target selection in Chinese reading. *Quarterly Journal of Experimental Psychology, 63*(4), 705–725. https://doi.org/10.1080/17470210903114858

Yan, M., Luo, Y., Xi, J., & Hao, Y. (2025). Parafoveal grouping of characters facilitates encoding of morphological hierarchical structure during sentence reading. *Reading and Writing*. https://doi.org/10.1007/s11145-025-10642-y

Yan, M., Pan, J., Chang, W., & Kliegl, R. (2019). Read sideways or not: Vertical saccade advantage in sentence reading. *Reading and Writing, 32*(8), 1911–1926. https://doi.org/10.1007/s11145-018-9930-x

Yan, M., Zhou, W., Shu, H., Yusupu, R., Miao, D., Krügel, A., & Kliegl, R. (2014). Eye movements guided by morphological structure: Evidence from the Uighur language. *Cognition, 132*(2), 181–215. https://doi.org/10.1016/j.cognition.2014.03.008